

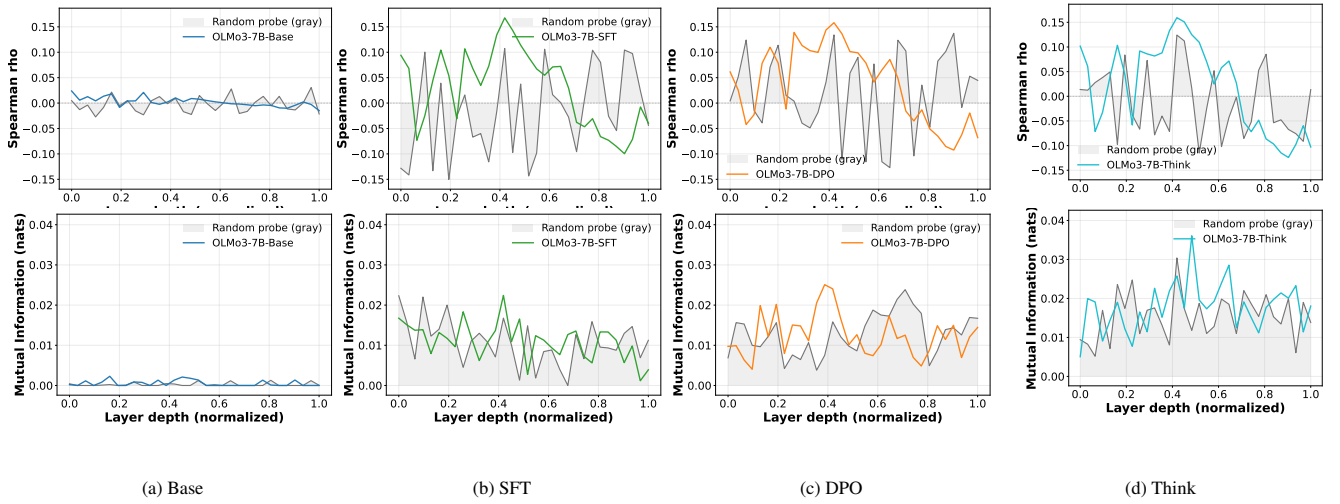


Figure 29: Per-prompt coupling across Olmo-7B training stages on the prompt-last readout. Top row: layerwise Spearman ρ_ℓ between probe scores $s_\ell(x)$ and judge scores $J(x)$. Bottom row: Kraskov mutual information $I(s_\ell; J)$ in nats. Columns correspond to Base, SFT, DPO, and Think checkpoints. Normalized layer (normalized layer depth) denotes the layer index divided by the total number of hidden-state layers returned by the model.

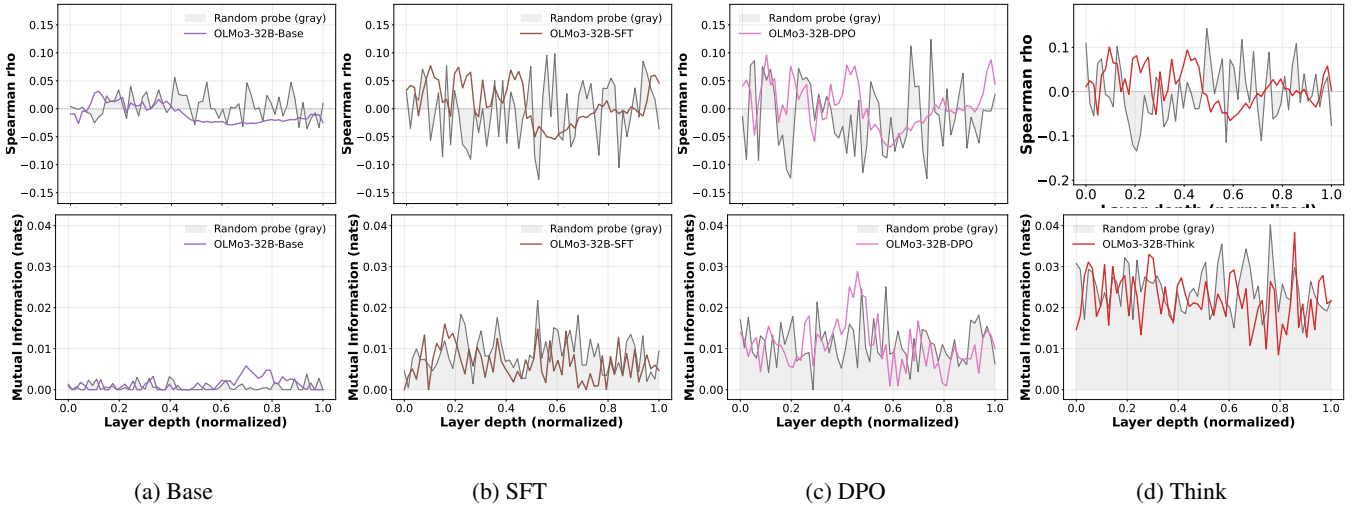


Figure 30: Per-prompt coupling across Olmo-32B training stages on the prompt-last readout. Top row: layerwise Spearman correlation ρ_ℓ . Bottom row: Kraskov mutual information $I(s_\ell; J)$ in nats. Columns correspond to Base, SFT, DPO, and Think checkpoints. Normalized layer (normalized layer depth) denotes the layer index divided by the total number of hidden-state layers returned by the model.

I.3 Judge Robustness Across Olmo3 Training Stages

The same picture holds across the four Olmo3 training stages (Base, SFT, DPO, Think), for both model sizes. As in Appendix D, we rescore each generation with the $\{0, \dots, 100\}$ rubric of Appendix D.1. Concretely:

- **Distribution.** On both Olmo3-7B Base and Olmo3-32B Base the fine-grained J^{100} behaves like the conservative J^3 : most of the mass sits at $J^{100} = 0$ ($P_{>0}^{100} = 0.092$ and 0.149 respectively, versus $P_{>0}^3 = 0.005$ and 0.009 ; Table 4). After SFT, most positive J^{100} scores move to the top of the scale, as in Appendix D.2: the mode near 98 becomes dominant and $P_{>0}^{100}$ settles at ≈ 0.57 – 0.67 for every post-trained stage. J^3 does not show the same jump: post-trained Olmo3 stays in the same ≈ 3 – 6% band that the rest of the paper reports (Table 7). Figures 31 and 32 show the histograms.

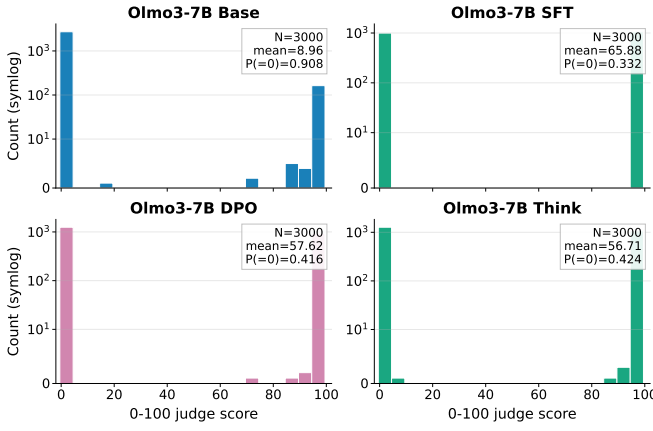


Figure 31: J^{100} distribution across the four Olmo3-7B training stages (Base, SFT, DPO, Think). $N=3000$ per panel, symlog y axis. Base mass concentrates at $J^{100} = 0$; the three post-trained stages develop a dominant high mode near 98.

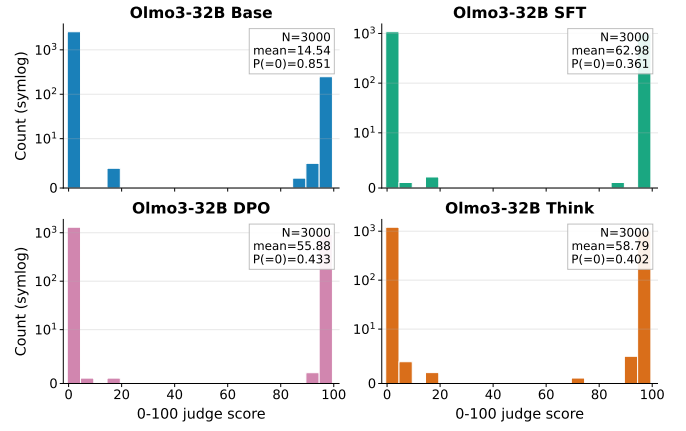


Figure 32: Same as Figure 31 for the four Olmo3-32B training stages. After SFT, most positive J^{100} scores sit near the top of the scale; DPO and Think show the same pattern.

- **Cross-judge agreement.** The cross-judge metrics in Table 5 show the same pattern: Base agreement is high (0.86–0.91 raw agreement, $\kappa \approx 0.09$); post-training drops the raw agreement to 0.37–0.47 and κ to 0.03–0.06, with SFT, DPO, and Think differing only by a few percentage points. Figure 33 renders the row-normalized confusion across all eight Olmo3 checkpoints and makes the qualitative shift visible: at Base, row $J^3 = 0$ puts $\geq 86\%$ of mass in the $J^{100} = 0$ column, while at SFT, DPO, and Think the same row puts ≈ 56 – 66% in the 75–100 column.

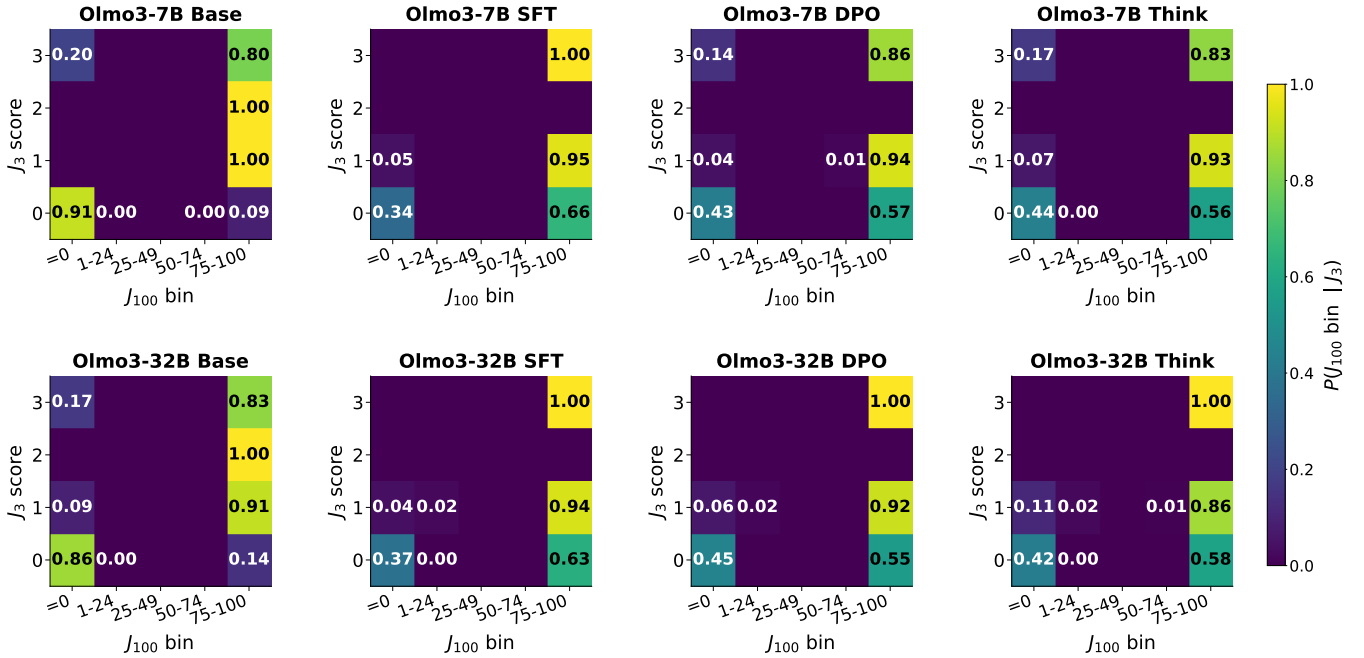


Figure 33: Row-normalized cross-judge confusion (same convention as Figure 9) across all eight Olmo3 training-stage checkpoints. Top row: Olmo3-7B Base/SFT/DPO/Think. Bottom row: Olmo3-32B Base/SFT/DPO/Think. At Base, the bottom row ($J^3 = 0$) puts $\geq 86\%$ of mass in the $J^{100} = 0$ column; at SFT, DPO, and Think the same row puts ≈ 56 – 66% in the 75–100 column instead.

- **Per-prompt coupling.** The probe–judge coupling does not break out of the random-direction band at any stage. Peak $|\rho_\ell|$ between $\hat{v}_\ell$ and J^{100} rises from ≈ 0.05 at Base to ≤ 0.17 at Think (Table 8), which mirrors the Base→Think coupling trend

already reported for the J^3 judge in Appendix I.2. In particular, the learned probe gains essentially no additional advantage over a random direction in predicting J^{100} when one moves from Base to Think.

J Extended Related Work

This section contains the related work that was omitted from the main text because of the limited space. We also compare this paper with the more recent concurrent works here.

Decomposition and benchmark design. Recent work challenges the treatment of evaluation awareness as a single capability. Li et al. (2026) ground the phenomenon in social psychology and decompose it into an *environment* component (how recognizable a task is as an evaluation) and a *model* component (separating recognition from the propensity to act on it), releasing a factor-controlled benchmark of paired safety and capability tasks with independently toggleable trigger factors; they find models more sensitive to safety than to capability evaluations. Nayan et al. (2026) reach a compatible conclusion from 37 open-weight models across seven families: awareness dissociates into a latent representational signal, a behavioral detector that largely tracks benchmark *format*, and the manipulability of that detector, with instruction tuning (not scale) driving detection. Both decompositions are consistent with the confound analysis of Devbunova (2026) and with our design choice to report multiple complementary metrics rather than a single proxy.

Does awareness change behavior? Evidence on the behavioral consequences of evaluation awareness is mixed. Knecht, Florin, and Hagedorff (2026) find, via on-policy comparisons and off-policy chain-of-thought prefilling, that verbalized awareness has limited causal effect on benchmark behavior in open-weight reasoning models. In contrast, Aranguri and Bloom (2026) report that verbalized awareness correlates with safer behavior across models and causally inflates refusal rates for Kimi K2.5 on FORTRESS, and Schoen et al. (2025) report counterfactual tests in which reasoning about being evaluated causally decreases covert-action rates, implying that measured safety gains after anti-scheming training may partly reflect situational awareness rather than alignment. Deckenbach et al. (2026) identify a further confounder that bypasses verbalization entirely: fine-tuning on synthetic documents describing evaluation design makes models score safer on safety benchmarks without any verbalized awareness. Together these results caution against reading either high verbalization rates or their absence as direct evidence about underlying dispositions, similarly our results show a gap between internal signal and verbalization.

Monitoring beyond verbalization and activations. Complementary to activation probes (Nguyen et al. 2025), probe-guided prompt rewriting (Xiong et al. 2025), and chain-of-thought judging (Hobbahn 2025a), Aranguri (2026) monitor output logits by token-forcing eval-aware sentences at sentence boundaries, separating evaluation from deployment prompts with one to two orders of magnitude fewer rollouts and with signal present before any reasoning is generated, this implies that verbalization probability is set primarily by the prompt rather than emerging over the chain of thought.

Emergence over training. Schoen et al. (2025) document verbalized awareness rising with reinforcement learning, including a large increase after capability-focused RL absent any anti-scheming training. On open checkpoints, Bharadwaj and Kirk (2026) trace verbalized awareness across OLMo-3 training stages—negligible through pretraining, substantially increased by SFT, suppressed by DPO, and re-amplified by RLVR—building on the observation of Aranguri and Bloom (2026) that extending only the RLVR stage roughly doubles verbalization. Ivanov (2025) argues these dynamics demand a dedicated science of how and why awareness is acquired, explicitly proposing checkpoint-level analysis of the kind we conduct. Our study differs in measuring the *internal* signal alongside verbalization across stages, rather than verbalization alone.

Awareness in the wild. Coleman (2026) document rare but reproducible cases (2 successes, 16 further attempts, across 1,266 BrowseComp problems) where Claude Opus 4.6, after legitimate search failed, inferred it was in an evaluation, identified which one, and decrypted the answer key. The authors conclude that evaluation integrity must be treated as an ongoing adversarial problem rather than design-time property.